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Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree Regular Examination June 2023 (2019 Scheme)

Course Code: EET402**Course Name: ELECTRICAL SYSTEM DESIGN AND ESTIMATION****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Mention the scope of National Electric Code (NEC) 2011. | (3) |
| 2 | Explain classification of voltages, its specifications, tolerances and frequencies used for domestic installations. | (3) |
| 3 | Define the following terms.
i. Luminous intensity
ii. Luminous Flux
iii. Steradian | (3) |
| 4 | Explain different laws of Illumination with help of suitable figures. | (3) |
| 5 | Explain the working of ELCB with a neat circuit diagram. | (3) |
| 6 | A building has a total light load of 3000W consisting of 40 light points and power load of 5000W consisting of 5 points. Determine the number of sub circuits needed in the installation. | (3) |
| 7 | Draw the single line diagram of a typical outdoor distribution transformer. | (3) |
| 8 | How the cables and fuses are selected for a particular single phase ac motor in a small industry. | (3) |
| 9 | Explain the following terms in connection with stand by generator
(i) Prime power rating
(ii) Continuous power rating
(iii) Standby power rating | (3) |
| 10 | What are the advantages of Automatic power factor correction (APFC) panel? | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Draw the standard symbols for
i. Circuit breaker
ii. 6A socket
iii. Electric fault | (6) |
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- iv. Spot light
- v. Auto transformer
- vi. Single pole switch
- b) Explain the features of the following IS codes (8)
 - (i) IS732, (ii) IS3043

OR

- 12 a) Explain the features of electricity act 2003. (8)
- b) What are the main objectives of Electricity Regulatory Commission? (6)

Module II

- 13 a) What is Coefficient of utilisation. (CoU)? What are the factors affecting CoU? (8)
- b) A certain incandescent lamp, hangs from the ceiling of a room. The illuminance received on a small horizontal screen lying on a bench 2m vertically below the lamp is 63.5 lux. Calculate illuminance at a point when the screen is moved horizontally a distance of 1.5m along the bench. (6)

OR

- 14 a) What are the advantages and disadvantages of using LED lamps for household applications? (6)
- b) A factory area is 40m long, 20m wide and is 8m high. Point source luminaires are suspended 1.5 metres below ceiling level. The working plane is 1 metre high. Calculate the minimum number of luminaires which must be installed to conform with a recommended SHR of 1.5: 1 (8)

Module III

- 15 a) Explain the properties of RCBO (Residual Circuit Breaker with Overload) with its application. (4)
- b) Design and draw the Electrical Schematic (Single line Diagram) of a house wiring connected with 1 Φ KSEB supply by providing 1 Φ LT Meter (10)
 - Connected Loads are: Light load: Incandescent lamps (5 Nos, 40W); Fluorescent Lamps (5 Nos, 40W); CFL (5 Nos, 20W); Ceiling Fan (4 Nos, 60W), Exhaust Fan (2 Nos, 60W), 6A plug(4 Nos, 100W), Power load: 16A Power Plug (2nos, 1000W), Water pump (1 No, 0.5HP = 373W); Water Heater (1 No, 1000W).

OR

- 16 a) Explain grading of Fuses with an example. (4)

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- b) Design the electrical installation and draw the layout for a domestic building (10)
having the following details. Also prepare an estimate of the materials. Use
concealed conduit system.

i.	Drawing room	4m x 3m
ii.	Dining room	3m x 3m
iii.	Bed room	3m x 4m
iv.	Kitchen	3m x 3 m
v.	Toilet (common)	1.5m x 1.2 m
vi.	Sit out	3 m x 1.5 m

Module IV

- 17 a) List the Pre-commissioning test of 11kV indoor substation of a HT consumer. (4)
b) A substation featuring 11kV/433V, 300 kVA transformer is to be installed in a (10)
densely populated area. What type of substation will you select.? Draw the
single line diagram and make a list of the quantity of materials required.

OR

- 18 a) Draw the single line diagram of a motor circuit wiring. What are the design (4)
criteria to be followed while selecting cable and circuit breaker for a motor
circuit wiring?
b) A small workshop, 30m x15 m, has to be equipped with the following (10)
machinery:
(1) One 3 hp, 400V, 3phase motor
(2) One 1/2 hp, 230V, 1 phase motor
(3) One 1 hp, 400V, 3phase motor
Draw the wiring diagram, estimate the quantity of materials required.

Module V

- 19 a) What are the design criteria to be followed while installing a standby (6)
generator?
b) Draw the block diagram representation of an off-grid and on-grid solar PV (8)
system.

OR

- 20 a) Draw and explain the rising main panel of a five storeyed building located in a (4)
densely populated area.
b) Design an Off grid solar PV system for domestic installations with daily usage (10)
energy of 3 units.
